HEXAGON

# Future of robotics 2035:

## Insights from industry experts

**autonomy**

*[ ɔːˈtɒnəmi ] : noun*
*the use of machines and computers that can operate without needing human control*

# The robotics renaissance is here

From ancient water clocks marking the hours to mobile robots directing complex warehouse logistics, autonomy has always been a testament to humanity's innate desire for progress. Its core promise endures. Increased productivity. Greater efficiency. Enhanced safety. Economic growth. All made possible by smart systems that streamline complex processes.

In 2025, we've reached a watershed moment. According to Forrester's Automation Survey, 2024, 82% of organisations have increased their investment in automation due to the **rise of generative AI** . Meanwhile, humanoid robots – absent from 2024's predictions – have now been named a top emerging technology.

Breakthroughs in AI, high-fidelity simulation, and multimodal sensing are pushing robots beyond scripted automation into the realm of embodied intelligence. Machines can increasingly perceive, learn, and independently make decisions in a variety of ways – and spaces – that were once uniquely human.

This begs the age-old question: where does that leave us?

As the technology curve dramatically steepens, evoking both sci-fi promise and dystopian unease, business leaders face a strategic imperative: capture the power of these innovations while safeguarding the human judgment, creativity, and purpose that define enduring organisations.

Autonomy has too often been treated as a zero-sum substitute for people. Instead, we must champion augmentation; designing a workforce where human ingenuity and robotic precision amplify one another to create value neither could achieve alone.

A robot renaissance is here. And with it, an extraordinary opportunity to reimagine how we produce and grow. This is our chance to build a truly intelligent future for robotics – a future where humans set the terms.

# Insights

We are standing on the shoulders of large language models to build a new generation of intelligent robots that can see, think, and act in the physical world – trained in simulation, refined with data, and optimised to run at the edge.

**Amit Goel, NVIDIA**

The leap from automation to autonomy hinges on a core shift in mindset. No longer are robots hard coded to follow rigid commands. Instead, they are rapidly trained in vast simulated environments until they achieve fluency in tasks. The real magic lies not in machines replacing us, but in how quickly they're learning to work with us.

**Burkhard Boeckem, Hexagon**

We're so focused on what AI can do that we forget to ask what it does to us. How does it change the way we think, write and interact? If we don't pay attention, we'll keep solving problems in ways that create new ones downstream. What we need is more humility, more intentionality, and a much more human-centred lens.

**Dr Juliet Aiken, Organisational Psychologist**

When we look at robotics, it's about freeing up people to do the 'soft, squishy brain' stuff where our people shine. A lot of what we do is look for repetitive work best suited for robots.

**Anthony Hartke, Turner Construction**

> AI needs a body and robotics needs AI.
>
> **Lukas Ziegler, Robotics Evangelist**

# Robotics and autonomy in 2025 – and beyond

## The robot economy is just getting started

Fast-forward from 1500 BCE to 2025. What once took centuries to invent is now evolving in mere months. As large language models (LLMs), digital twins, simulation, and multimodal sensor technologies converge, a new breed of autonomous agents is arising with the dexterity to thread fibre-optic cable, the judgement to reroute around a fallen pallet, and the anatomy to service a satellite placed in low-Earth orbit.

Meanwhile, the economic case for robotics grows stronger by the day. Adoption is accelerating where labour is scarcest. More than 50 countries are watching their working-age populations contract. In the United States alone, manufacturers face a **1.9-million-person skills gap** by 2033.

In response, e-commerce leaders have deployed upwards of three-quarters of a million mobile and articulated robots, unlocking double-digit productivity gains in logistics. Thousands of last-mile deliveries are made by drones while autonomous inspection units roam mines and offshore rigs.

And humanoids? They are slowly edging their way into industrial workplaces for pick-and-place, maintenance, and disaster-recovery tasks.

Now, Asia-Pacific – home to 70% of new industrial robot installations – stands to reshape global competitiveness as nations **race to embed adaptive autonomy**. And with worldwide robotics revenue projected to reach US$50.80bn this year, the commercial and political **stakes have never been higher**.

Today, the definition of autonomy is changing. It's no longer about simply extracting humans from workflows but about designing systems around us – around our environments, behaviours, and purpose. It's about working together to strengthen industrial resilience, enable workforce adaptability, and redefine economic prosperity.

# Insights

Over the next five years, we'll see major growth in co-bot deployment, as well as AI-powered autonomy that enables robots to work in far more dynamic, unpredictable environments. With real-time learning and edge computing, we'll unlock new possibilities in outdoor robotics, urban navigation, and time-sensitive industrial automation.

**Chase Williams, TechInsights**

What's happening now is that robots are no longer just static, purpose-built machines. With transformers, vision-language models, and LLMs, we now have foundation models with a strong semantic understanding of the world. They may not know about their own body yet, but they know a lot about how the world operates.

**Amit Goel, NVIDIA**

Robots won't replace humans for complex tasks anytime soon, but they're already proving valuable by automating repetitive, injury-prone work. This shift enables smarter allocation of human talent toward higher-value roles. Every task we automate frees up capacity elsewhere, reshaping opportunity cost across the workforce.

**Spencer Huang, NVIDIA**

The real issue behind all this is demographic collapse. Automation isn't optional. It's the only way forward. It's how we keep systems running, industries moving, and quality of life intact. We're not talking about replacing people, we're talking about sustaining economies that would otherwise grind to a halt.

**Mario Mauerer, maxon Group**

We often think of robots and humans either doing their own thing and combining results later or passing tasks down a line. But real collaboration means solving problems together, in the same space, and dynamically responding to each other. Perhaps if we start thinking about robots as teammates instead of tools, we open up possibilities for healthier, more flexible work relationships.

**Dr Juliet Aiken, Organisational Psychologist**

Gen Z aren't exactly lining up to work in warehouses or chemical plants. They want to be digital nomads, content creators, freelancers. And yet, this same generation expects fast fashion, one-day shipping, on-demand everything. So, the demand is growing, but the labour force is shrinking. That's not a threat. It's a gap that needs filling.

**Lukas Ziegler, Robotics Evangelist**

There's this idea that a humanoid will walk into a factory and do everything a human can, perfectly, straight away. We're not there. And that's okay.

What we're seeing in the near term is robots being deployed in the places where they make the most sense; handling the dangerous jobs, the repetitive, tedious tasks that wear people down or put them at risk. Those are areas where robots can already deliver real value, right now. As they become more capable? We'll expand from there.

**Burkhard Boeckem, Hexagon**

Spatial intelligence enables humanoids to perceive, adapt, and act in various environments. Innovation in sensor fusion and Visual Language Action (VLAs) models is necessary to drive progress in this field.

**Arnaud Robert, Hexagon**

# Key challenges and barriers to adoption

## Breaking the negative feedback loop

But where there are opportunities, there are also challenges. The real world is complicated, messy, and unpredictable – especially where humans are involved. In our passionate pursuit of progress, we risk becoming architects of a hollow future, automating for automation's sake without a clear understanding of its impact. As intelligence advances, issues across technical, commercial, psychological, and ethical domains are becoming entangled and creating a negative feedback loop. Engineering limits inflate costs; cost pressures heighten scrutiny; scrutiny feeds public anxiety; and societal anxiety fuels tighter governance, which in turn raises the technical bar.

# Challenges

## Technical

Power-hungry actuators, patchy perception in unconstrained spaces, and the lack of large-scale real-world training data can leave even the most state-of-the-art robots struggling with open-ended reasoning or unexpected edge cases.

**Insights**

Robots require more than just perception; they need to understand the physical world and reason about how to interact with it. That means capturing data not just of how things look, but of how they behave when touched, moved, or resisted. This is the next major hill to climb: building a baseline corpus of manipulation data that can serve the same purpose for robotics that books and blogs did for LLMs.

**Spencer Huang, NVIDIA**

There's this idea that if we want robots to operate in human spaces, they need to look and move exactly like humans. But robots don't need 27-degree-of-freedom hands or two identical arms just because we have them. What robots really need are smart, modular end effectors designed specifically for the job. A simple gripper that picks up a box. A soft-touch mechanism that handles fragile components. Maybe even a pointy bit that presses buttons.

**Mario Mauerer, maxon Group**

We humans tend to think linearly. Technology doesn't. While breakthroughs like ChatGPT have conditioned us to expect going from zero to 80% functionality relatively quickly, the final 10–20% – the leap from good to exceptional – is where progress slows dramatically. This friction at the frontier is not a failure, it's simply a reality of physics, complexity, and integration.

**Lukas Ziegler, Robotics Evangelist**

## Commercial

High CapEx, integration with legacy systems, cybersecurity overheads, and unclear ROI make boards reluctant to move past pilots, especially for SMEs that can't write off costs at scale.

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The biggest barriers to robotics adoption are cost, regulation, and cybersecurity. High upfront costs and unclear ROI make it hard for SMEs to justify the investment, especially when installation, training, and maintenance are often underestimated. On top of that, there's no universal safety standard, which makes liability a major concern.

**Chase Williams, TechInsights**

Companies often fixate on the physical robot (its dexterity, its sensor stack, its AI) but the real complexity lies in everything that surrounds it. Questions like: who is responsible for day-to-day operations? How do we train staff? How does the robot talk to our legacy systems? Where does the data go? Who owns it? Who maintains the robot, and how often? These aren't edge cases. They're core to successful adoption.

**Mario Mauerer, maxon Group**

What works best is early collaboration, bringing us in while vendors are still developing solutions. That's how you make sure the tool fits into real workflows on site, rather than us having to reverse-engineer value from a finished product. When we get to help shape the tech, that's when adoption takes off.

**Anthony Hartke, Turner Construction**

## Commercial continued

There's a lot of hype around full autonomy but what's often overlooked is that most of the real commercial value today comes from simpler, more targeted applications. In many industrial settings, the challenge isn't finding a robot that can think like a human, it's solving a very specific problem reliably. Maybe it's pressing a reset button. Maybe it's delivering a part from A to B. The best solution often isn't the most advanced one. It's the most practical one.

**Burkhard Boeckem, Hexagon**

Industrial environments demand near-perfection. A robot with 99% pick-and-place accuracy may seem impressive, but in logistics this isn't good enough. That's because the industrial environments robots must operate in require near-flawless execution to be both safe and productive. Businesses are chasing 99.999999% – where every fraction of a percentage point matters.

**Lukas Ziegler, Robotics Evangelist**

We don't have the luxury of just building a new plant tomorrow. Any new technology we introduce has to be integratable into our existing, operational facilities. It has to fit into brownfield environments. And beyond that, it needs to deliver on the current KPIs that already matter in real production: reliability, repeatability, and time-to-market. That's the only way we can justify it from an ROI perspective.

**Expert in future robotics, working at a leading German manufacturer**

## Psychological

When robotic systems are designed with human needs in mind, they are more likely to be welcomed with curiosity, enthusiasm, and even pride. Yet, persistent psychological frictions, from the 'uncanny valley' discomfort of human-like machines to the mental fatigue of constant high-level decision-making, can just as easily undermine long-term success.

**Insights**

When we talk about automation, we often celebrate the removal of repetitive tasks, but we forget that those tasks serve a purpose. They give our brains room to breathe. What I see with AI and robotics is a growing tendency to strip jobs of their natural rhythm: either leaving workers with relentless cognitive strain or dull, unthinking machine babysitting. Both extremes are terrible for mental health.

**Dr Juliet Aiken, Organisational Psychologist**

Reactions among our staff to having robot colleagues have been very positive. You walk a site and see guys showing off the robot to visitors, to each other, taking videos, posting on social media. The prevailing vibe is: 'I don't totally know what it does yet, but it looks cool, and I want to be around it'.

**Anthony Hartke, Turner Construction**

What we've seen from our workforce is genuine curiosity. Successful integration doesn't hinge only on hardware or software; it depends on creating a cultural environment where people are open to working alongside machines. That openness among our employees has been encouraging to witness.

**Expert in future robotics, working at a leading German manufacturer**

People expect robots to cook dinner, carry shopping bags, and do the laundry. But we're still facing serious hardware challenges. There's a risk here: if robots don't live up to the hype, people will write them off altogether. They'll say, "Remember the robot revolution? That didn't work out." And they'll miss the real progress being made in systems that don't look like us but are quietly transforming industries.

**Lukas Ziegler, Robotics Evangelist**

## Governance

Executives will continue to find themselves navigating a regulatory and reputational minefield. Data privacy, algorithmic bias, safety standards, and the urgent issue of widening labour inequities must be thoughtfully managed.

**Insights**

Automation tends to reflect the values it's built with – efficiency, standardisation, scale. But those aren't inherently human values. When you design only for output, you risk building systems that are profitable but soulless. We often talk about AI replacing people as if humans are just faulty machines. But the value of human work isn't just in the outcome, it's in the process, the decisions, the experience.

**Dr Juliet Aiken, Organisational Psychologist**

At some point, a robot will cause an accident. What matters is not whether mistakes happen, but how quickly and transparently we learn from them. Safety, in this context, becomes a kind of social contract that balances trust, regulation, and continuous improvement.

**Mario Mauerer, maxon Group**

Many businesses realise they need a shift in design philosophy in a lot of cases, once they see how poorly designed automation leads to user confusion and overreliance. And that risk translates directly into robotics when systems aren't really built with human context in mind.

**Chase Williams, TechInsights**

In our work, there's no tolerance for risk when it comes to data or connectivity. You can't just take a cloud-connected robot still in development and throw it on a secure site. These systems have to be able to operate offline on their own, and still meet government-grade security standards. That's non-negotiable.

**Anthony Hartke, Turner Construction**

Safety is absolutely paramount for both our people and our data, especially because humans and robots work side by side in our factories. When it comes to humanoids, the challenge is even bigger. There are no established ISO safety standards yet. We can't just let vendors walk in and train their AI on our shop floor without clear compliance frameworks. That's a major barrier to scaling these technologies in the real world.

**Expert in future robotics, working at a leading German manufacturer**

We tend to overestimate the effect of a technology in the short run and underestimate it in the long run.

Amara's Law

# 2035: snapshots of an autonomous future

Roy Amara reminds us that today's hype often masks tomorrow's deeper transformations. By 2035, autonomy won't just drive cars or patrol factory floors, it will thread itself into the machinations of work, the pulse of cities, and even our routines in subtle yet decisive ways.

These snapshots aren't science fiction; they're signals already visible in today's deployments, quietly laying the foundations for a future where autonomy doesn't just change how we work, but what work is.

Crucially, without careful design, autonomy risks entrenching existing labour divides by concentrating high-prestige, cognitively rich roles among the privileged, while relegating others to low-agency, low-reward tasks.

To confront this, we must together rethink our economic, labour, and production models, ensuring the future of work is not just technologically advanced, but fair, sustainable, and human-centred.

# Three possible worlds

1

## A robot revolution

- Robots handle the majority of repetitive physical work across Tier-1 manufacturing, logistics, and frontline services
- Human labour pivots to design, oversight, and high-empathy roles
- Sovereign wealth funds compete for 'robot dividends'
- Government-funded skills credit schemes pay citizens to learn new tech
- Humanoid total cost of ownership (TCO) is cheaper than emerging-market wages

2

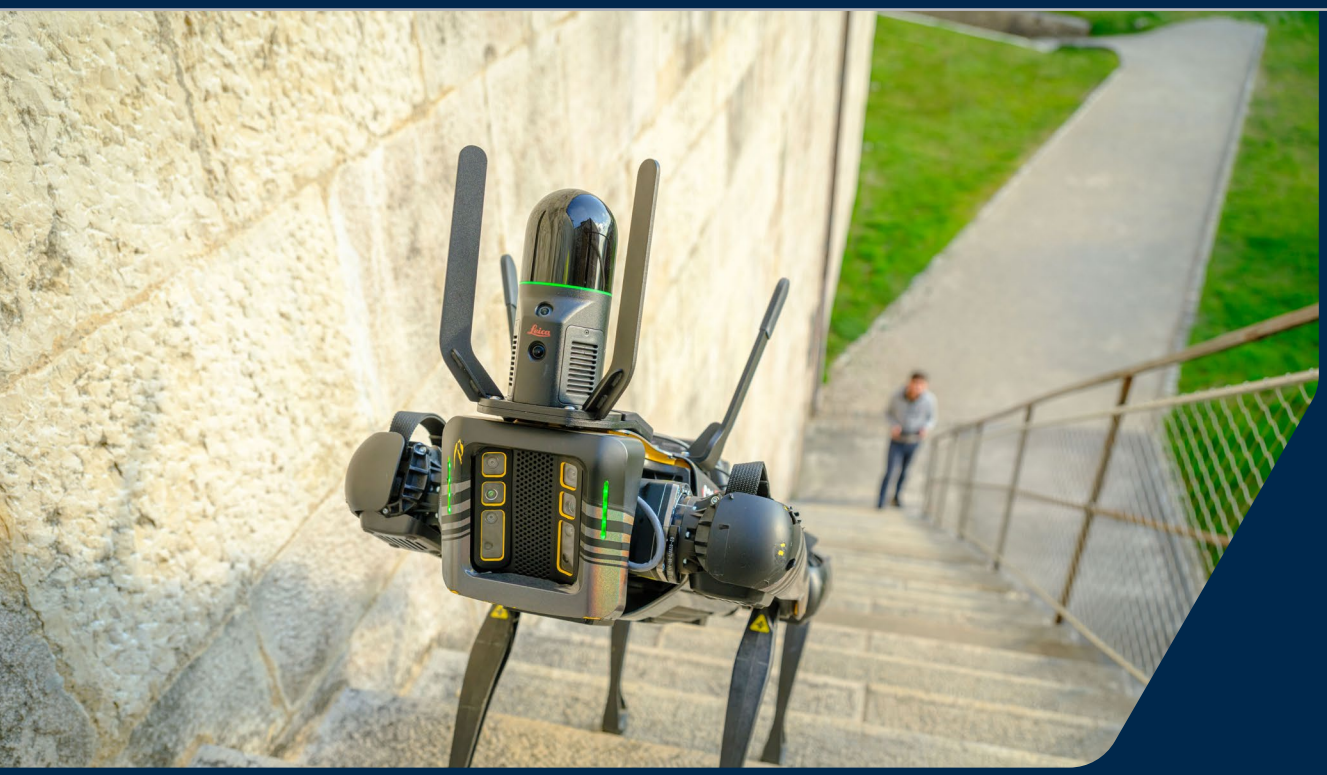
## The cautionary tale

- Edge-AI stalls under data-locality mandates
- Public trust erodes as workers displaced with nowhere to go
- Cyber-sabotage and safety incidents trigger prohibitions
- Governments impose 'robot licensing', capping deployment density
- A factory accident and a surveillance scandal ignite a 'Neo-Luddite' movement

3

## A managed mesh (most likely)

- Around half of industrial and service tasks are robotised
- Fortune 500 companies carry a robotics line-item in OpEx budget.
- Robots occupy predictable, semi-structured roles such as e-commerce fulfilment, urban maintenance, and healthcare assistance
- Industry-wide Robot-as-a-Service API standard enables plug-and-play robotics
- Annual cyber-physical stress tests mandatory after high-profile supply-chain hack



# What we want 2035 to look like

Shaping a better future calls for more than data. What we need is imagination. The following visions for our world in ten years, shared by those building, using, and studying autonomous systems today, offer glimpses of where we could be headed – if we can get things right.

In 2035, robots are an invisible infrastructure layer, every bit as common as a smartphone today. Factory floors are lights-out with a handful of people steering fleets of machines, while pop-up micro-factories made of mobile, reconfigurable robot cells let even tiny teams turn out custom products on demand. Real-time digital twins trial every tweak before metal moves, slashing waste, and edge AI will make split-second calls on-site without risking data privacy.

# Future blueprints

Round-the-clock warehouses and swarms of ground and aerial couriers use predictive control and dynamic stock placement to make online shopping cheaper, faster, and lower-emission than ever. Falling hardware prices and open-source AI models will spread these gains to factories and farms everywhere.

**Chase Williams, TechInsights**

In 2035, you can walk into a store or log into a platform and buy a robot that fits your specific environment. Whether it's a nimble assistant for a hospital corridor or a compact unit for your home, the hardware is no longer the barrier. Even better, you don't need a PhD or a decade of engineering experience to make it useful. You simply tell it what you want – through language, gestures, or demonstration – and it learns to fold your towels the way you want them folded.

This level of flexibility and personalisation is only possible because the robotics ecosystem embraces openness. Instead of hoarding data and locking down platforms, developers around the world contribute to a shared knowledge base, something like a global effort to teach robots how to navigate, manipulate, and reason about the world.

**Spencer Huang, NVIDIA**

Done right, I could see a future where automation helps to rebuild the middle class. Expensive college degrees have been replaced by technician pathways, which you can step into straight out of high school and grow in, rather than going broke for a diploma that doesn't teach you practical skills. By embracing apprenticeships, rethinking job design, and investing in training, we create jobs that are truly tenable. Without intention, though, automation could exacerbate existing socio-economic differences.

**Dr Juliet Aiken, organisational psychologist**

In 2035, every crew has its own logistics robot, something that's not flashy or humanoid, but genuinely useful. A kind of pack mule that knows how to move materials, tools, drywall, whatever's needed, between points on site. You tell it, 'Go to room 503,' and it does. Or it follows you around with what you need. If something goes wrong, you ask it what's wrong and it tells you.

**Anthony Hartke, Turner Construction**

In 2035, task-unified machines that can do more than one thing at a time make up a portion of our workforce. Imagine a bot that scans for floor flatness, checks on workers' safety, and helps finish a floor, all in a single trip. We're getting real ROI with a reliable, rugged, mobile teammate that shows up to work every day and makes everyone's life easier.

Kristofer Wahl, Turner Construction

By 2035 robots won't require a PhD to operate. In industries where there just aren't enough people, robots will arrive with built-in intelligence. You'll be able to take one out of the box, bring it into a factory or warehouse, and show it what to do through simple demonstrations or prompts.

**Amit Goel, NVIDIA**

In 2035, humanoid robots are a natural part of the industrial landscape. They walk the same factory floor we do – opening doors, climbing stairs, operating equipment – because their form has been designed to fit the environment. But what really sets them apart is their artificial intelligence.

These aren't pre-programmed machines. They learn by imitating, adapt to new tasks within hours, and handle edge cases with real-world awareness. Robots don't just automate work, they understand it. And that makes them indispensable.

**Burkhard Boeckem, Hexagon**

Robots are solving real problems, not chasing overpromised vision. We see mobile manipulation becoming standard in environments where it delivers real business value. There are scaled deployments where robotics is either saving money or making money, tied to strong use cases, not hype. Just intelligent, human-aware machines doing their jobs, so humans can focus on theirs.

**Lukas Ziegler, robotics evangelist**

## How do we steer towards the future we all hope for? The decade ahead will be shaped by a series of pivotal choices: where we invest, which standards we adopt, how boldly we regulate, and how inclusively we bring people along.

We must see autonomy not as a substitute for people, but as a scaffold that amplifies human capability and widens opportunity.

That means investing first in skills, designing every system for transparency and dignity, and measuring success by shared prosperity, not throughput alone. It means legislators partnering with industry to embed ethics into code, and boards rewarding long-term resilience over quarterly gains. It means governments creating the guardrails for responsible innovation and businesses being held accountable for contributing to the public good.

Above all, it means inviting workers, communities, and under-represented voices to co-create this next industrial era.

### Design for dignity

Embed inclusive, accessible UX from day one.

Build requirements, wireframes, and safety cases with frontline users, rather than simply delivering to them.

### Deploy with empathy

Pilot in real workplaces, closing live feedback loops.

Robots enter production only through phased pilots that surface friction, trust, and governance gaps.

### Evolve with trust

Continuously refine systems through transparent metrics.

Adoption is treated as a living relationship – updates, reskilling, and governance iterate together.

# Insights

Embracing Robotics-as-a-Service could convert heavy CapEx into a steady, predictable OpEx stream. By tracking metrics like cycle time, defect rate, uptime, and throughput right from day one, we'll know exactly where the gains are. Interoperability, too, has to be non-negotiable, ensuring new robots plug easily into our existing systems, while robust onboarding and hands-on training equip people to work alongside the technology.

**Chase Williams, TechInsights**

The roadmap to scale lies in democratisation – making it possible to train and re-task robots using natural language, visual inputs, or intuitive interfaces. Robotics must be treated as a global open-source mission. Everyone has something different to contribute, whether that's data, design, hardware, intelligence. Walling it off slows progress for everyone. The fastest path forward is shared.

**Spencer Huang, NVIDIA**

Robotics is a massive systems problem. You need innovation across hardware, sensors, simulation, AI models, rendering, control – at least 50 different technologies have to come together. There's no way any one company can solve all of it alone. That's why building an ecosystem is essential.

**Amit Goel, NVIDIA**

The brain needs variety to thrive. When we strip that away, we may be making tasks easier, but we're making the jobs themselves unbearable. I'm not saying efficiency is bad, but it can't be the only value that drives design. We need to be brave enough to ask: what kind of work feels meaningful, not just productive?

**Dr Juliet Aiken, Organisational Psychologist**

Change management becomes the hidden architecture of robotics success. In many cases, the first real bottleneck isn't the robot's capabilities. It's the company's readiness to adapt.

To move beyond pilots and scale to meaningful deployment, businesses must approach robotics as an enterprise transformation project, not a procurement decision.

**Mario Mauerer, maxon Group**

Let's shift the narrative. Robots aren't displacing people; they're helping businesses stay afloat in a world that's fundamentally changing. When designed responsibly, they're creating independence, stability, and continuity not only for industry, but for society as a whole. In this, I believe humanoids should play a key role. But only where they make economic and functional sense. They should be used because they solve a clear, immediate problem not because they look cool in a press release. If we keep our focus there, we'll move robotics forward in a way that lasts.

**Lukas Ziegler, Robotics Evangelist**

When people work side by side with machines, they need to feel safe, comfortable, and in control. It sounds simple, but it influences everything from the materials we use to how we think about form, motion, and safety. You don't want a 2-tonne, 2-metre robot marching around. We take inspiration from aerospace and Formula 1 to reduce mass and energy use, building systems that can move safely and intuitively in shared spaces.

**Burkhard Boeckem, Hexagon**

> Advances in robotics and artificial intelligence are opening many new opportunities – a key challenge is to ensure they translate into real business value.

**Arnaud Robert, Hexagon**

# Hexagon: building an autonomous future

## Engineering empathy into autonomy

For Hexagon, the answer lies in a deeply human-centric vision of progress. The purpose of autonomy should be to augment human capabilities, to solve the workforce crises of tomorrow, and to create industries that are not only more efficient but more humane.

Hexagon didn't start with humanoids. We started with motorised surveying instruments that allowed humans to stand where they wanted to measure instead of being stuck behind a tool. That was robotics in a way. And that idea – putting intelligence into the machine to make life easier for people – has been with us since the beginning.

Robots that operate in harmony with people. AI that adapts to our ways of working. For us, this continues to be an active design principle. Through breakthrough innovations in humanoid robotics and intelligent sensing, we are engineering technologies that respect the social, psychological, and economic dimensions of the environments they stand to enhance.

This isn't about science fiction. It's about building systems that do something useful. Sometimes full autonomy is more than you need. A robot that can press a button, so a human doesn't have to. Sometimes you want more – to expand what's possible with a new generation of autonomous systems that feel less like machines and more like trusted collaborators. With deep, longstanding relationships with customers across the globe, Hexagon understands the nuance of industrial workflows and is designing robots that integrate into real systems.

Our focus on integration, not spectacle, ensures we are ready to meet society's most urgent industrial challenges and reclaim a bolder, brighter, and more intelligent future for all.